

From Asymmetric TD-Error Bias to Counterfactual Action-Advantage Geometry

Mathematical Development of the Research Question

1 Where the project started

The starting observation was empirical: asymmetric manipulation of the temporal-difference (TD) error produced systematic behavioral changes in a PPO agent.

For the original binary feeding problem, let

$$a \in \{E, S\},$$

where E denotes eating and S denotes skipping.

The empirical observation was approximately

$$\boxed{\text{positive TD clipping} \longrightarrow \text{under-eating}}$$

and

$$\boxed{\text{negative TD clipping} \longrightarrow \text{over-eating.}}$$

The first theoretical question was therefore:

Why should clipping only one side of the TD-error distribution systematically alter the learned behavior?

Answering this question eventually leads to a more general question:

What determines the *direction* in which an advantage perturbation moves a policy?

This document derives that progression.

2 Step 1: TD error as an advantage estimator

For a policy π , define

$$V^\pi(s) = \mathbb{E}_\pi[G_t \mid s_t = s]$$

and

$$Q^\pi(s, a) = \mathbb{E}_\pi[G_t \mid s_t = s, a_t = a].$$

The advantage is

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s).$$

Assume initially an exact critic:

$$V_\phi(s) = V^\pi(s).$$

The TD residual is

$$\delta_t = r_t + \gamma V^\pi(s_{t+1}) - V^\pi(s_t).$$

Conditioning on $(s_t = s, a_t = a)$ gives

$$\mathbb{E}[\delta_t \mid s, a] = \mathbb{E}[r_t + \gamma V^\pi(s_{t+1}) \mid s, a] - V^\pi(s) \quad (1)$$

$$= Q^\pi(s, a) - V^\pi(s) \quad (2)$$

$$= \boxed{A^\pi(s, a)}. \quad (3)$$

Thus, under the exact-critic assumption,

$$\boxed{\mathbb{E}[\delta_t \mid s, a] = A^\pi(s, a)}.$$

For idealized infinite-horizon GAE,

$$\hat{A}_t = \sum_{\ell=0}^{\infty} (\gamma\lambda)^\ell \delta_{t+\ell}.$$

For $\ell \geq 1$,

$$\mathbb{E}[\delta_{t+\ell} \mid s_{t+\ell}] = \sum_a \pi(a \mid s_{t+\ell}) A^\pi(s_{t+\ell}, a) = 0.$$

Hence the future residuals vanish in conditional expectation and

$$\boxed{\mathbb{E}[\hat{A}_t \mid s_t = s, a_t = a] = A^\pi(s, a)}.$$

This establishes the clean reference signal before perturbation.

3 Step 2: positive clipping creates a downward advantage bias

Let positive TD errors be clipped at $c_+ > 0$:

$$\tilde{\delta}_t^+ = \min(\delta_t, c_+).$$

Write this as

$$\tilde{\delta}_t^+ = \delta_t - h_+(\delta_t),$$

where

$$h_+(\delta) = (\delta - c_+)_+ = \max(\delta - c_+, 0).$$

The corresponding clipped GAE is

$$\tilde{A}_t^+ = \sum_{\ell=0}^{\infty} (\gamma\lambda)^\ell \tilde{\delta}_{t+\ell}^+.$$

Define the action-conditional positive-tail removal

$$B^+(s, a) = \sum_{\ell=0}^{\infty} (\gamma\lambda)^\ell \mathbb{E}[(\delta_{t+\ell} - c_+)_+ \mid s_t = s, a_t = a].$$

Since

$$B^+(s, a) \geq 0,$$

we obtain

$$\mathbb{E}[\tilde{A}_t^+ \mid s, a] = A^\pi(s, a) - B^+(s, a).$$

Therefore positive clipping creates a non-positive pointwise bias.

However, this result alone does *not* explain under-eating.

The missing question is:

$$\text{Is the bias different across actions?}$$

4 Step 3: negative clipping creates an upward advantage bias

Similarly, let negative TD errors be clipped at $c_- < 0$:

$$\tilde{\delta}_t^- = \max(\delta_t, c_-).$$

This can be written

$$\tilde{\delta}_t^- = \delta_t + h_-(\delta_t),$$

where

$$h_-(\delta) = (c_- - \delta)_+.$$

Define

$$B^-(s, a) = \sum_{\ell=0}^{\infty} (\gamma\lambda)^\ell \mathbb{E}[(c_- - \delta_{t+\ell})_+ \mid s_t = s, a_t = a].$$

Then

$$\mathbb{E}[\tilde{A}_t^- \mid s, a] = A^\pi(s, a) + B^-(s, a).$$

Thus negative clipping creates a non-negative pointwise bias.

Again, this does not by itself imply over-eating.

Behavior depends on the *relative action-dependent bias*.

5 Step 4: the first important realization — relative action advantage matters

For every state,

$$V^\pi(s) = \sum_a \pi(a \mid s) Q^\pi(s, a).$$

Therefore

$$\sum_a \pi(a \mid s) A^\pi(s, a) = 0.$$

For the binary eat/skip policy, define

$$p(s) = \pi(E \mid s), \quad 1 - p(s) = \pi(S \mid s).$$

Then

$$pA_E + (1 - p)A_S = 0.$$

The behaviorally relevant quantity is therefore not merely A_E or A_S separately, but

$$A_E - A_S.$$

For positive clipping,

$$\begin{aligned}\tilde{A}_E^+ &= A_E - B_E^+, \\ \tilde{A}_S^+ &= A_S - B_S^+.\end{aligned}$$

Hence

$$\tilde{A}_E^+ - \tilde{A}_S^+ = (A_E - A_S) - (B_E^+ - B_S^+). \quad (4)$$

If

$$\boxed{B_E^+ > B_S^+},$$

then

$$\boxed{\tilde{A}_E^+ - \tilde{A}_S^+ < A_E - A_S}.$$

Thus positive clipping reduces the relative preference for eating.
Likewise, for negative clipping,

$$\tilde{A}_E^- - \tilde{A}_S^- = (A_E - A_S) + (B_E^- - B_S^-).$$

If

$$\boxed{B_E^- > B_S^-},$$

then

$$\boxed{\tilde{A}_E^- - \tilde{A}_S^- > A_E - A_S}.$$

Thus negative clipping increases the relative preference for eating.
This gives the original behavioral mechanism:

$$\boxed{\text{action-dependent TD tails} \rightarrow \text{action-dependent advantage bias} \rightarrow \text{relative action-preference change.}}$$

6 Step 5: connect relative advantage to the policy gradient

The policy gradient is

$$\nabla_{\theta} J = \mathbb{E}[A(s, a) \nabla_{\theta} \log \pi_{\theta}(a | s)].$$

For the binary policy, the state-conditional actor gradient can be reduced to

$$\boxed{g(s) = \nabla_{\theta} p(s) (A_E - A_S)}.$$

Therefore positive clipping gives

$$g_+(s) = \nabla_{\theta} p [A_E - A_S - (B_E^+ - B_S^+)].$$

The gradient bias is

$$\boxed{g_+(s) - g(s) = -\nabla_{\theta} p (B_E^+ - B_S^+)}.$$

If

$$\nabla_{\theta} p > 0 \quad \text{and} \quad B_E^+ > B_S^+,$$

then

$$g_+(s) - g(s) < 0.$$

Positive clipping therefore suppresses gradient pressure toward eating.

Similarly,

$$g_-(s) - g(s) = +\nabla_\theta p(B_E^- - B_S^-).$$

If

$$\nabla_\theta p > 0 \quad \text{and} \quad B_E^- > B_S^-,$$

negative clipping increases gradient pressure toward eating.

This established that the observed feeding effect is fundamentally a *directional policy-gradient effect*.

7 Step 6: the binary proof exposes the more general problem

The binary result depends on

$$A_E - A_S.$$

But this immediately raises a problem for a general action space

$$\mathcal{A} = \{a_1, \dots, a_K\}.$$

For $K > 2$, there is no unique scalar difference such as

$$A_E - A_S.$$

Instead, the natural object is

$$\mathbf{A}(s) = \begin{bmatrix} A(s, a_1) \\ A(s, a_2) \\ \vdots \\ A(s, a_K) \end{bmatrix} \in \mathbb{R}^K.$$

The original binary difference was therefore only a special projection of a more general *action-preference vector*.

This is the first conceptual transition:

$$A_E - A_S \implies A(s, \cdot).$$

8 Step 7: generalize clipping bias to arbitrary action spaces

Let

$$z_\theta(s, a) = \nabla_\theta \log \pi_\theta(a \mid s)$$

be the policy score.

At a fixed state,

$$g(s) = \mathbb{E}_{a \sim \pi}[A(s, a)z_\theta(s, a)].$$

The score identity gives

$$\mathbb{E}_{a \sim \pi}[z_\theta(s, a)] = 0.$$

Under positive clipping,

$$\tilde{A}^+(s, a) = A(s, a) - B^+(s, a).$$

Therefore

$$g_+(s) = \mathbb{E}_a[(A - B^+)z_\theta] \quad (5)$$

$$= g(s) - \mathbb{E}_a[B^+(s, a)z_\theta(s, a)]. \quad (6)$$

Hence

$$\Delta g_+(s) = -\mathbb{E}_a[B^+(s, a)\nabla_\theta \log \pi(a | s)].$$

Likewise,

$$\Delta g_-(s) = +\mathbb{E}_a[B^-(s, a)\nabla_\theta \log \pi(a | s)].$$

This is the generalized form of the original clipping theorem.

The central causal chain becomes

$$\text{action-dependent perturbation} \rightarrow \text{score-weighted gradient bias} \rightarrow \text{directional policy change}.$$

9 Step 8: why a state-only advantage bias is insufficient

Suppose we perturb every action at state s by the same scalar $c(s)$:

$$\tilde{A}(s, a) = A(s, a) + c(s).$$

The additional policy gradient is

$$\Delta g(s) = \mathbb{E}_a[c(s)\nabla_\theta \log \pi(a | s)] \quad (7)$$

$$= c(s)\mathbb{E}_a[\nabla_\theta \log \pi(a | s)] \quad (8)$$

$$= \boxed{0}. \quad (9)$$

Therefore

$$\text{an action-independent advantage offset cannot create a first-order policy push.}$$

This was the crucial bridge from “bias magnitude” to “bias direction.”

What matters is not simply

how much the advantage is changed,

but

$$\text{how the perturbation varies across actions at the same state.}$$

10 Step 9: define a general steering vector

Let

$$\mathbf{d}(s) = [d_1(s), \dots, d_K(s)]^\top$$

represent a desired action-preference direction.

Remove its policy-weighted constant component:

$$\bar{d}_\pi(s) = \sum_a \pi(a | s) d_a(s).$$

Define the centered perturbation

$$f(s, a) = d_a(s) - \bar{d}_\pi(s).$$

Then

$$\sum_a \pi(a | s) f(s, a) = 0.$$

The modified advantage is

$$\tilde{A}(s, a) = \hat{A}(s, a) + \alpha f(s, a),$$

where α controls perturbation strength.

The additional gradient is

$$\Delta g = \alpha \mathbb{E}[f(s, a) \nabla_\theta \log \pi(a | s)].$$

Thus the original clipping bias can now be viewed as one particular mechanism for generating an action-dependent direction field.

This is the second conceptual transition:

$$\text{uncontrolled clipping-induced bias} \implies \text{controlled action-space steering.}$$

11 Step 10: categorical policy geometry

Consider a categorical softmax policy

$$\pi = \text{softmax}(\mathbf{z}),$$

where $\mathbf{z} \in \mathbb{R}^K$ are policy logits.

The softmax Jacobian is

$$C_\pi = \text{diag}(\pi) - \pi \pi^\top.$$

For an action-direction vector $\mathbf{d}$, the first-order logit push has the form

$$\Delta \mathbf{z} \propto C_\pi \mathbf{d}.$$

This equation provides the geometric interpretation that was missing from the original clipping proof.

The raw vector

$$\mathbf{d}$$

specifies which action preferences we wish to raise or lower, while

$$C_\pi$$

determines how much of that direction the current policy can actually express.

Hence:

$$\text{steering direction} = \text{action-preference direction, not state-space direction.}$$

12 Step 11: what should the steering direction be?

Once the intervention is written as a vector over actions, the next question is:

What is the principled direction to use?

The natural candidate is the policy's own full action-advantage vector:

$$\mathbf{d}(s) = \mathbf{A}(s, \cdot).$$

Because

$$\sum_a \pi(a | s) A(s, a) = 0,$$

the advantage vector is already policy-centered.

An aligned intervention is

$$\mathbf{d}_{\text{good}}(s) = +\mathbf{A}(s, \cdot).$$

An opposed intervention is

$$\mathbf{d}_{\text{bad}}(s) = -\mathbf{A}(s, \cdot).$$

Conceptually,

$$\hat{A} + \alpha A$$

reinforces the existing learning direction, whereas

$$\hat{A} - \alpha A$$

suppresses it and, for sufficiently large α , can reverse it.

The original positive/negative clipping phenomenon has therefore become a more general question:

What happens when we deliberately change the direction of the action-advantage signal?

13 Step 12: the counterfactual problem

Standard PPO trajectories reveal only the advantage associated with the sampled action:

$$(s_t, a_t, \hat{A}_t).$$

But the steering vector requires

$$\mathbf{A}(s) = [A(s, a_1), \dots, A(s, a_K)]^\top.$$

Therefore we need information about actions that were *not* taken.

This produces the counterfactual question:

At the same state, what would have happened under every alternative action?

This is how the bias-perturbation proof leads naturally to counterfactual action evaluation.

14 Step 13: explicit counterfactual action landscape

At a saved state s , restore the simulator and evaluate every action:

$$a_1, \dots, a_K.$$

For each action,

$$\hat{Q}_{\text{CF}}(s, a) = r(s, a) + \gamma V(s').$$

Define the policy-weighted measured baseline

$$\bar{V}_{\text{CF}}(s) = \sum_a \pi(a | s) \hat{Q}_{\text{CF}}(s, a).$$

Then

$$A_{\text{CF}}(s, a) = \hat{Q}_{\text{CF}}(s, a) - \bar{V}_{\text{CF}}(s).$$

By construction,

$$\sum_a \pi(a | s) A_{\text{CF}}(s, a) = 0.$$

The explicit counterfactual procedure therefore gives exactly the object suggested by the generalized clipping proof:

$$s \longrightarrow \mathbf{A}_{\text{CF}}(s, \cdot).$$

15 Step 14: from measured vectors to a state-conditioned direction field

Explicit counterfactual evaluation is expensive.

If the action landscape is structured by state, we can instead learn

$$B_\phi(s) \approx \mathbf{A}_{\text{CF}}(s, \cdot).$$

For linear features $\psi(s)$,

$$B_\phi(s) = B^\top \psi(s).$$

Given

$$\Psi = \begin{bmatrix} \psi(s_1)^\top \\ \vdots \\ \psi(s_N)^\top \end{bmatrix}$$

and

$$Y = \begin{bmatrix} \mathbf{A}_{\text{CF}}(s_1)^\top \\ \vdots \\ \mathbf{A}_{\text{CF}}(s_N)^\top \end{bmatrix},$$

ridge regression gives

$$B = (\Psi^\top \Psi + \lambda I)^{-1} \Psi^\top Y.$$

Thus we obtain a state-conditioned action-direction field

$$s \longrightarrow B_\phi(s) \in \mathbb{R}^K.$$

16 Step 15: why trajectory-derived counterfactuals are the next step

The explicit simulator construction solves the conceptual problem, but it introduces additional environment interactions.

That raises the next question:

Can the counterfactual action landscape be inferred from the trajectories PPO already collects?

This motivates a trajectory-based counterfactual estimator, such as a hindsight/counterfactual credit-assignment method:

$$\boxed{\text{PPO trajectories} \longrightarrow A_{\text{traj-CF}}(s, \cdot).}$$

The explicit simulator counterfactual is then retained as an oracle:

$$\boxed{A_{\text{CF}}^{\text{explicit}} \stackrel{?}{\approx} A_{\text{traj-CF}}.}$$

We then amortize the trajectory-derived landscape:

$$\boxed{B_\phi(s) \approx A_{\text{traj-CF}}(s, \cdot).}$$

The full progression is therefore

$$\boxed{\text{PPO trajectories} \rightarrow A_{\text{traj-CF}} \rightarrow B_\phi(s) \rightarrow \text{directional PPO intervention},}$$

with

$$A_{\text{CF}}^{\text{explicit}}$$

used as a validation reference.

17 Step 16: the current PPO intervention

Ordinary PPO uses

$$\hat{A}_{\text{GAE}}(s, a).$$

The simplest counterfactual intervention is

$$\boxed{\hat{A}_{\text{new}}(s, a) = \hat{A}_{\text{GAE}}(s, a) + \alpha B_\phi(s)_a.}$$

Consequently,

$$g_{\text{new}} = \mathbb{E}[\hat{A}_{\text{new}} \nabla_\theta \log \pi(a \mid s)] \tag{10}$$

$$= g_{\text{PPO}} + \alpha \mathbb{E}[B_\phi(s)_a \nabla_\theta \log \pi(a \mid s)]. \tag{11}$$

Define

$$g_{\text{CF}} = \mathbb{E}[B_\phi(s)_a \nabla_\theta \log \pi(a \mid s)].$$

Then

$$\boxed{g_{\text{new}} = g_{\text{PPO}} + \alpha g_{\text{CF}}.}$$

This is the current experimental mechanism.

18 Step 17: from bias to angle

The original work asked whether a perturbation pushes behavior toward eating or skipping.

The generalized framework asks for the *angle* between learning directions.

For a categorical policy, define the policy-induced inner product

$$\langle x, y \rangle_\pi = x^\top C_\pi y.$$

The corresponding norm is

$$\|x\|_\pi = \sqrt{x^\top C_\pi x}.$$

For the reference advantage direction A and intervention direction d ,

$$\cos \theta_\pi(A, d) = \frac{A^\top C_\pi d}{\sqrt{A^\top C_\pi A} \sqrt{d^\top C_\pi d}}.$$

This gives

$$\theta = 0^\circ : \quad \text{aligned,}$$

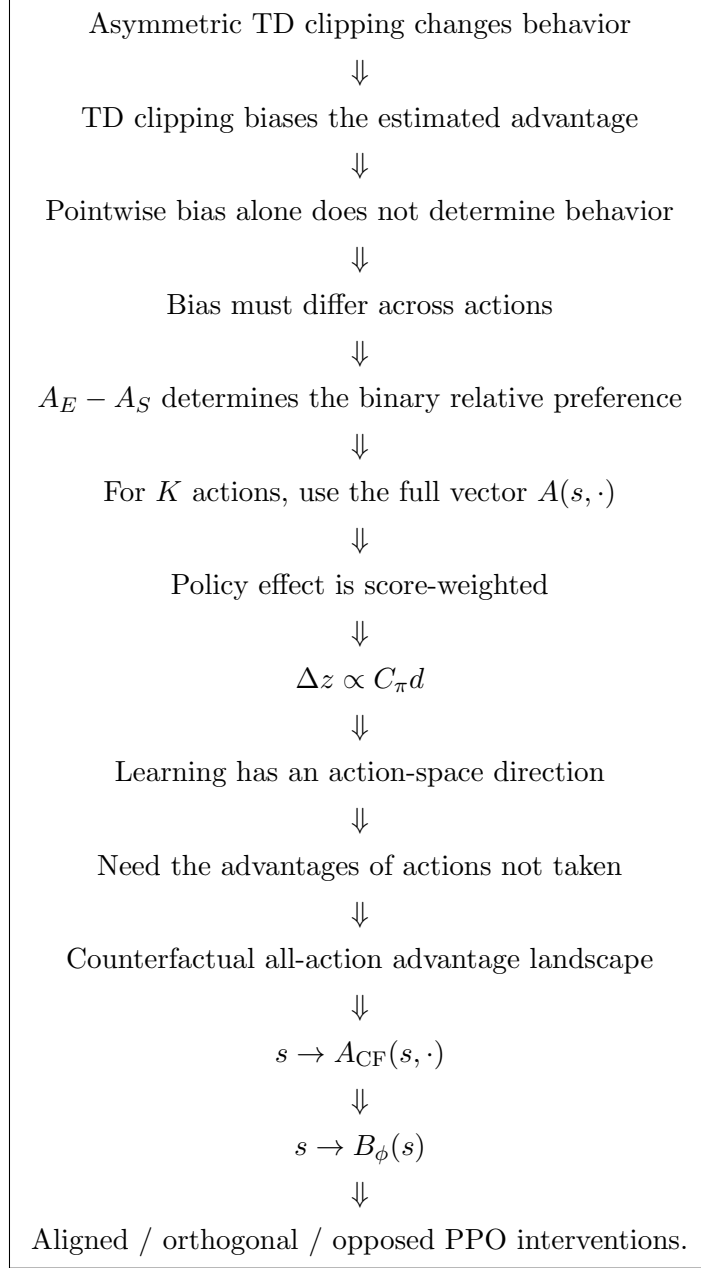
$$\theta = 90^\circ : \quad \text{orthogonal,}$$

$$\theta = 180^\circ : \quad \text{opposed.}$$

Thus the original binary “increase/decrease eating” result becomes a continuous geometric intervention framework.

19 Step 18: the complete conceptual chain

The research development can now be written in one sequence:



20 Current scientific hypothesis

The current central hypothesis can therefore be stated as:

Policy-gradient learning is governed by a state-dependent direction in action-preference space. Asymmetric TD perturbations reveal that changing this direction changes behavior. If the full counterfactual action-advantage direction can be recovered from trajectories and predicted at unseen states, then its alignment with the PPO update can be deliberately controlled, allowing systematic manipulation—and potentially improvement—of learning dynamics.

More formally, let g_{oracle} denote the gradient direction induced by the explicit counterfactual action landscape.

We test whether

$$\cos(g_{\text{new}}, g_{\text{oracle}}) > \cos(g_{\text{PPO}}, g_{\text{oracle}})$$

for an aligned intervention.

If improved alignment is behaviorally meaningful, we further expect

better directional alignment $\longrightarrow$ faster and/or more stable policy learning.

The corresponding negative test is equally important:

opposed direction $\longrightarrow$ degraded or reversed learning pressure.

21 Why this is a generalization rather than a separate idea

The counterfactual action-advantage framework is not disconnected from the original clipping result.

The original clipping proof identified the essential mechanism:

behavioral change arises from action-dependent distortion of the learning signal.

The current framework removes the dependence on a specific clipping rule and asks for the general object responsible for that effect:

the state-dependent action-space learning direction.

Thus:

asymmetric clipping $\subset$ action-dependent advantage perturbations $\subset$ directional policy-gradient interventions.

Counterfactual estimation is then the mechanism that allows us to observe the full direction rather than only the component corresponding to the sampled action.

22 Important qualification

The derivation above describes the first-order actor signal.

PPO additionally uses the clipped surrogate objective

$$L^{\text{CLIP}} = \mathbb{E}[\min(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t)],$$

where

$$\rho_t = \frac{\pi_{\theta}(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}.$$

At the beginning of an update,

$$\theta = \theta_{\text{old}}, \quad \rho_t = 1,$$

so the derivations describe the initial actor-gradient direction.

They do not guarantee the exact final finite-step policy after multiple epochs, minibatches, PPO ratio clipping, critic approximation, and changes in the state distribution.

This is why the empirical study must measure both

gradient direction

and

actual policy/return evolution.

23 One-line research pitch

Observed bias $\rightarrow$ action-dependent advantage distortion $\rightarrow$ policy-gradient direction $\rightarrow$ counterfactual action geo

We begin from the observation that asymmetric TD-error perturbations systematically bias PPO behavior, show that the effect arises from action-dependent distortion of the policy-gradient signal, generalize the binary relative-advantage mechanism to a full action-space geometry, and use counterfactual action advantages to measure and deliberately control the direction of policy learning.